

Protection afforded by post-infection SARS-CoV-2 vaccine doses: a cohort study in Shanghai

Reviewed Preprint

v2 • September 24, 2024

Revised by authors

Reviewed Preprint

v1 • April 26, 2024

Bo Zheng, Bronner Gonçalves, Pengfei Deng, Weibing Wang, Jie Tian, Xueyao Liang, Ye Yao , Caoyi Xue 

Department of Epidemiology, School of Public Health, Fudan University, Shanghai 200032, China • Department of Comparative Biomedical Sciences, School of Veterinary Medicine, University of Surrey, Guildford, UK • Shanghai Pudong New Area Center for Disease Control and Prevention, Shanghai, 200120, China • Key Laboratory of Public Health Safety of Ministry of Education, Fudan University, Shanghai 200032, China • School of Public Health, Shanghai Institute of Infectious Disease and Biosecurity, Fudan University, Shanghai 200032, China

 https://en.wikipedia.org/wiki/Open_access

 Copyright information

Abstract

Background

In many settings, a large fraction of the population has both been vaccinated against and infected by SARS-CoV-2. Hence, quantifying the protection provided by post-infection vaccination has become critical for policy. We aimed to estimate the protective effect against SARS-CoV-2 reinfection of an additional vaccine dose after an initial Omicron variant infection.

Methods

We report a retrospective, population-based cohort study performed in Shanghai, China, using electronic databases with information on SARS-CoV-2 infections and vaccination history. We compared reinfection incidence by post-infection vaccination status in individuals initially infected during the April-May 2022 Omicron variant surge in Shanghai and who had been vaccinated before that period. Cox models were fit to estimate adjusted hazard ratios (aHR).

Results

275,896 individuals were diagnosed with RT-PCR-confirmed SARS-CoV-2 infection in April-May 2022; 199,312/275,896 were included in analyses on the effect of a post-infection vaccine dose. Post-infection vaccination provided protection against reinfection (aHR 0.82; 95% CI 0.79-0.85). For patients who had received one, two or three vaccine doses before their first infection, hazard ratios for the post-infection vaccination effect were 0.84 (0.76-0.93), 0.87 (0.83-0.90) and 0.96 (0.74-1.23), respectively. Post-infection vaccination within 30 and 90 days before the second Omicron wave provided different degrees of protection (in aHR): 0.51 (0.44-

0.58), and 0.67 (0.61-0.74), respectively. Moreover, for all vaccine types, but to different extents, a post-infection dose given to individuals who were fully vaccinated before first infection was protective.

Conclusions

In previously vaccinated and infected individuals, an additional vaccine dose provided protection against Omicron variant reinfection. These observations will inform future policy decisions on COVID-19 vaccination in China and other countries.

eLife assessment

This **important** work by Zheng and colleagues uses a large cohort database from Shanghai to identify that post-infection vaccination among previously vaccinated individuals provides significant low to moderate protection against re-infection. The evidence supporting the conclusion is **convincing** with some limitations, e.g., lack of symptom severity as an outcome, and no inclusion of time since infection as an independent variable). This study will be of interest to vaccinologists, public health officials and clinicians.

<https://doi.org/10.7554/eLife.94990.2.sa3>

1 Introduction

Four years after the first reports of severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) infection, the Coronavirus disease (COVID-19) pandemic continues to be a global concern, especially due to the risk of emergence of new variants[1, 2]. In most countries, the variant that is currently epidemiologically dominant is the Omicron[3, 4], which, due to its increased transmissibility and high number of mutations, led to significant increases in the number of infections in 2022[5]. Omicron variant infections were first observed in China in December 2021[3], and in Shanghai, the spread of the Omicron BA.2 sublineage led to a substantial increase in COVID-19 incidence between February 26 and June 30, 2022[6].

In December 2022[7], an important change in the COVID-19 policy in China, namely the end of most social distancing measures and of mass screening activities, was associated with a second surge in SARS-CoV-2 infections in Shanghai. The current circulation of the virus in the Shanghaiese population and reports of vaccine fatigue mean that it is important to estimate the protective effect of vaccination against reinfection in this population. In this study, we aimed to quantify the effect of vaccine doses given after a first infection on the risk of subsequent infection. For that, we used data collected during the first Omicron variant wave, when hundreds of thousands of individuals tested real-time polymerase chain reaction (RT-PCR)-positive for SARS-CoV-2 infection[8] in Shanghai, of which 275,896 individuals in Pudong District. The fact that the population in Shanghai was mostly SARS-CoV-2 infection naïve before the spread of the Omicron variant provides a unique opportunity to estimate the real-world benefit of post-infection vaccine doses in a population that was first exposed to infection during a relatively short and well-defined time window. We further investigated whether the number of pre-infection vaccination doses modified the protective effect of the post-infection dose against Omicron BA.5 sublineage. To avoid ambiguity in the text, in the following sections, we often refer to vaccine doses given after the initial infection as “post-infection vaccination” or “post-infection vaccine dose”.

2 Results

2.1 Demographic characteristics of the study population

Of the 275,896 individuals with RT-PCR-confirmed SARS-CoV-2 infection during the first Omicron variant wave (from April 1 to May 31, 2022) in Pudong, Shanghai, 1,227 individuals died from reasons unrelated to COVID-19 between the first infection and November 2022 and were excluded from our analysis. Most first infections (243,906, 88.8%) occurred in April; for 306 (0.1%) individuals, information on the date of first infection was not available. In April 2022, more than half of the study population (68.6%) had completed full vaccination and one third (34.4%) had received booster vaccination.

To assess the effect of post-infection vaccination, the analytic sample consisted of 199,312 individuals ([Figure 1](#)). 85,804 were women (43.1%); 836 (0.4%) had sex information missing. 38.1% of the study participants were aged 20 to 39 years and only 0.9% were aged 0 to 6 years (see [Table 1](#) and [Table S1](#) for additional information).

2.2 Vaccination after first Omicron variant infection

[Figure 2](#) (Panel A) shows vaccination coverage in the analytic population over time. In April 2022, a series of emergency epidemic prevention measures, including mass screening with Nucleic Acid Amplification Tests (NAATs), city-wide lockdown, home quarantine of residents, were implemented in Shanghai, leading to the temporary suspension of vaccination in April and May 2022. By the end of the study period, January 2023, 69.5% (190,815/274,363) of the study participants had completed full vaccination, and 38.4% booster vaccination ([Figure S1](#)). Vaccination coverages for the first, second, third, and fourth vaccine doses were 72.7%, 67.6%, 37.4% and 0.3%, respectively.

As mentioned above, only participants who had received vaccination before the first infection were included in this analysis. After infection, 10,241 (5.1%), 5,096 (2.6%), and 810 (0.4%) individuals received another vaccine dose in August, and during the periods from September to November 2022 and from December 2022 to January 2023, respectively. Between their first infection, in April-May 2022, and January 2023, 17.4% (2,466/14,131) of the study participants who had one pre-infection dose received a second dose of COVID-19 vaccine, and 13.8% (12,886/93,087) of those who had received two pre-infection doses received a third vaccine dose. Only 1.0% (142/14,131) of the study participants who had one pre-infection dose received two post-infection vaccine doses, and 0.2% (144/93,087) of those who had received two pre-infection doses received two post-infection doses ([Table S2](#)). All individuals with three pre-infection vaccine doses who received a fourth dose (795/92,094, 0.9%) received the post-infection dose in December 2022; for 793 of these participants, the post-infection vaccination received in December was their second booster vaccination (for the other 2, their fourth dose was the first booster vaccination; see [Supplementary Appendix](#) for more information about vaccine policy in Shanghai).

2.3 SARS-CoV-2 reinfections

Among the study participants, 48,651/199,312 (24.4%) had SARS-CoV-2 reinfection. The median age of individuals with no evidence of reinfection was 41.7 years (IQR: 31.0, 55.7 years), and of individuals with confirmed SARS-CoV-2 reinfection, 45.8 years (IQR: 34.0, 55.9 years) ([Table S1](#)). The median time interval between the first infection and reinfection was 244.7 days (IQR: 237.4, 250.0), which implies that the risk of misclassifying a long infection as a reinfection was low.

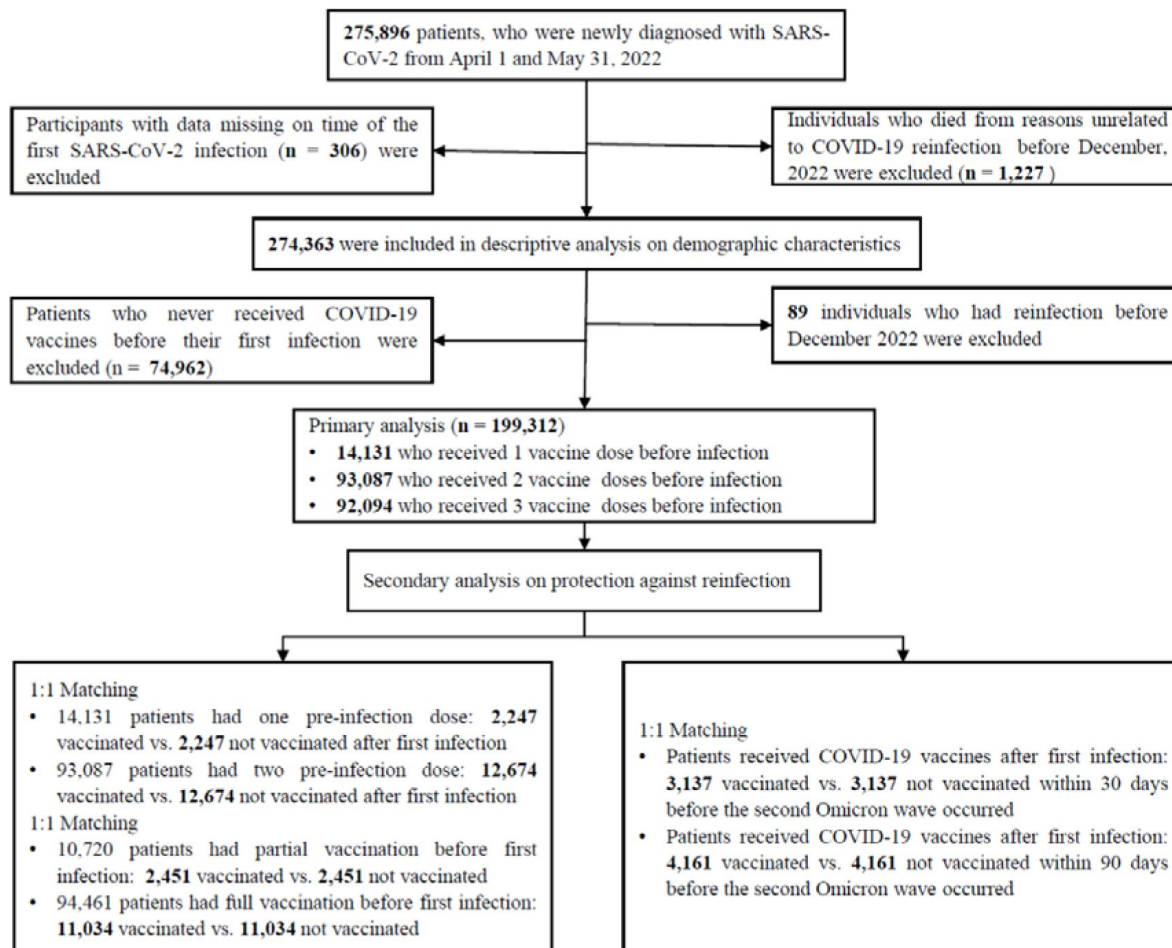


Figure 1.

Flow chart describing the selection of participants for the analysis.

The number of individuals in this figure is not the same as some of the numbers in Table 1 because of missing data in key variables. Note that in the bottom part of the chart, related to secondary analyses, the boxes represent overlapping sets of study participants; in other words, some individuals included in the secondary analyses that correspond to the left box were also included in analyses corresponding to the box on the right.

Characteristics		All		No post-infection vaccination		Post-infection vaccination	
		N (%)	Reinfection rate, % (95% CI).	N (%)	Reinfection rate, % (95% CI).	N (%)	Reinfection rate, % (95% CI).
Overall		199,312	24.4 (24.2, 24.6)	183,165	24.7 (24.5, 24.9)	16,147	21.5 (20.8, 22.2)
Sex	Male	112,672 (56.5)	26.1 (25.8, 26.4)	104,002 (56.8)	26.4 (26.1, 26.7)	8,670 (53.7)	22.3 (21.3, 23.3)
	Female	85,804 (43.1)	22.4 (22.1, 22.7)	78,403 (42.8)	22.9 (22.6, 23.2)	7,401 (45.8)	17.4 (16.4, 18.3)
Age, years	0-6	1,736 (0.9)	7.0 (5.8, 8.3)	1,569 (0.9)	6.6 (5.4, 8.0)	167 (1.0)	10.2 (6.2, 15.9)
	7-19	10,762 (5.4)	13.0 (12.3, 13.7)	10,347 (5.6)	12.9 (12.3, 13.6)	415 (2.6)	14.7 (11.4, 18.8)
	20-39	75,955 (38.1)	22.4 (22.1, 22.8)	71,005 (38.8)	22.7 (22.3, 23.0)	4,950 (30.7)	19.1 (17.9, 20.3)
	40-59	74,680 (37.5)	29.4 (29.0, 29.8)	70,569 (38.5)	29.6 (29.2, 30.0)	4,111 (25.5)	25.8 (24.2, 27.4)
	60+	35,903 (18.0)	22.6 (22.1, 23.1)	29,446 (16.1)	23.7 (23.1, 24.2)	6,457 (40.0)	17.6 (16.6, 18.6)
Regions	Shanghai	44,259 (22.2)	18.9 (18.5, 19.3)	41,250 (22.5)	19.4 (19.0, 19.9)	3,009 (18.6)	11.9 (10.7, 13.1)
	Other provinces	44,959 (22.6)	20.9 (20.5, 21.3)	43,045 (23.5)	21.0 (20.6, 21.5)	1,914 (11.9)	18.2 (16.4, 20.2)
Occupations	Preschoolers and students	12,232 (6.1)	12.1 (11.5, 12.7)	11,677 (6.4)	12.0 (11.4, 12.6)	555 (3.4)	13.2 (10.4, 16.4)
	Employed	29,537 (14.8)	20.8 (20.2, 21.3)	28,343 (15.5)	20.8 (20.3, 21.4)	1,194 (7.4)	18.8 (16.4, 21.3)
	Retired	37,482 (18.8)	22.5 (22.0, 23.0)	30,955 (16.9)	23.5 (23.0, 24.1)	6,527 (40.4)	17.5 (16.5, 18.5)
	Working age not in labor [†]	5,606 (2.8)	21.0 (19.8, 22.2)	5,311 (2.9)	21.5 (20.3, 22.8)	295 (1.8)	11.5 (8.1, 15.9)
Clinical severity*	Asymptomatic	81,584 (40.9)	19.9 (19.6, 20.2)	77,057 (42.1)	20.3 (19.9, 20.6)	4,527 (28.0)	14.1 (13.1, 15.3)
	Mild/moderate	7,602 (3.8)	19.9 (18.9, 21.0)	7,216 (3.9)	20.1 (19.1, 21.2)	386 (2.4)	16.6 (12.9, 21.0)
	Severe or critical	32 (0.0)	15.6 (5.9, 34.3)	22 (0.0)	13.6 (3.8, 36.4)	10 (0.1)	20.0 (4.0, 64.1)

CI: confidence interval; [†] people of working age (≥18 years) unemployed or not in the labor force (disabled); *Clinical severity of first infection.

Table 1.

Characteristics of the study population and reinfection rate by post-infection vaccination status.

Here, reinfection rate refers to the percentage of the relevant study subpopulation with evidence of reinfection between December 1, 2022 and January 3, 2023. Note that for the variables on region, occupation, and clinical severity, data are missing for large fractions of the study population. Note also that information was only available on sex at birth, but not on gender.

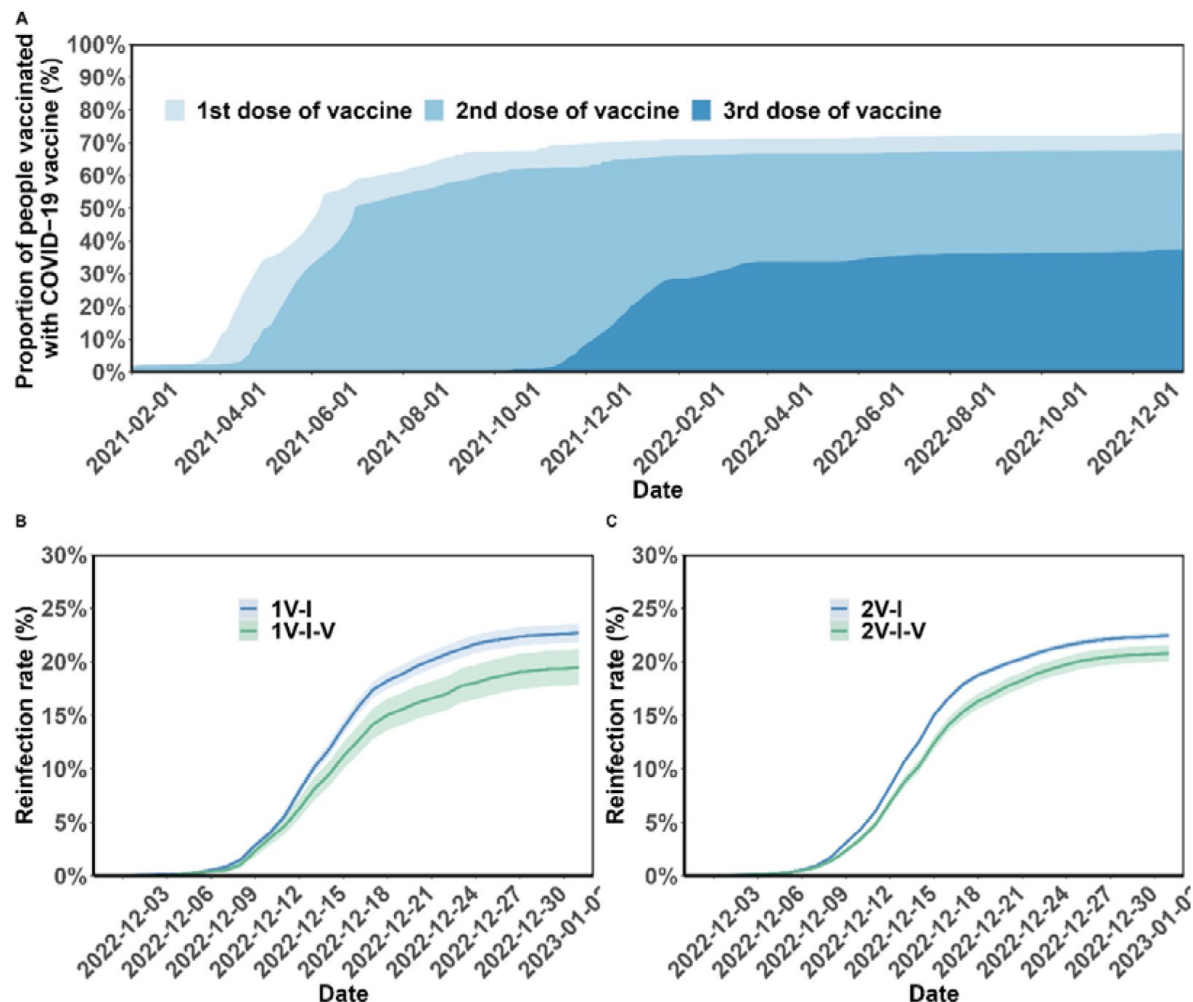


Figure 2.

Vaccination coverage and cumulative incidence of SARS-CoV-2 reinfection in the study population.

Panel A presents the percentages of the study population vaccinated over time. The cumulative incidence of SARS-CoV-2 reinfections is presented by the number of vaccination doses before (panels) and after (lines) first infection (panels B-C). Shaded regions: 95% CIs. 1V-I, 2V-I represented 1, and 2 vaccine doses before infection, respectively; 1V-I-V, 2V-I-V corresponds to 1 and 2 doses before infection, then post-infection vaccination, respectively. As mentioned in the *Results* section, 142 and 144 study participants who received one and two pre-infection vaccine doses received two post-infection vaccine doses. We do not show the corresponding plot for those individuals who received 3 pre-infection doses as their post-infection dose was after the start of the follow-up, in December.

Figure 2 (Panels B and C) shows the cumulative incidence of SARS-CoV-2 reinfection by vaccination status. Overall, individuals who were not vaccinated after their first infection were more often reinfected compared to those who received post-infection vaccination. The percentage of female individuals who became reinfected was lower than that of male individuals (22.4% vs. 26.1%; **Table 1**); and reinfection was more common in participants aged 40 to 59 years compared to those aged 20 to 39 (29.4% vs. 22.4%). Individuals originally from other provinces had a slightly higher risk of reinfection compared to individuals from Shanghai (20.9% vs. 18.9%). The risks for retired individuals, for those of working age who were not working, and for those working were 22.5, 21.0, and 20.8%, respectively.

2.4 Vaccine effectiveness

For individuals who had received at least one pre-infection vaccine dose, post-infection vaccination was protective against reinfection (adjusted hazard ratio [aHR] 0.82, 95% CI 0.79, 0.85). As shown in **Figure 3**, this protective effect was observed in subgroups defined by the number of pre-infection vaccine doses: aHR of 0.84 (95% CI, 0.76, 0.93) and 0.87 (95% CI, 0.83, 0.90) for one and two pre-infection doses respectively; and for patients with three vaccine doses prior to infection, the association was not statistically significant (aHR: 0.96 [0.74, 1.23]). When analyses are stratified by partial and full vaccination status before the first infection, post-infection vaccination was protective (aHR 0.76 [0.68, 0.84], and 0.93 [0.89, 0.97], respectively); and among individuals who had received booster vaccination before the first Omicron variant wave in Shanghai, the hazard ratio estimate was consistent with a more limited effect (aHR: 0.95 [0.75, 1.22]). Note that the overlapping categories between classification of pre-infection vaccination status and the number of pre-infection vaccinations (**Table S3**). For comparison, results for individuals who had not been vaccinated before their first infection are shown in the *Supplementary Appendix* (supplementary section “Effect of post-infection vaccination in individuals with no history of vaccination before infection” and **Table S4**).

In analyses stratified by demographic characteristics (**Figure 4**), post-infection vaccine doses were protective in both female (aHR 0.81 [0.76, 0.86]) and male individuals (aHR 0.83 [0.79, 0.87]). Post-infection vaccination was more protective for participants aged 60 years or older (aHR 0.73 [0.69, 0.78]) compared to other age groups. The estimated aHR for individuals who were asymptomatic during their first infection was 0.80 (0.74, 0.87), and for those who were symptomatic during the first infection was 1.01 (0.78, 1.29).

As a secondary analysis, we estimated vaccine effects by calendar time of vaccination. Post-infection vaccine doses given within 30 and 90 days of the second Omicron variant wave were associated with lower hazard of reinfection (aHRs 0.51 [0.44, 0.58], 0.67 [0.61, 0.74], respectively). Note that in this secondary analysis, exposed and unexposed individuals were matched using propensity scores. We also performed a secondary analysis by vaccine type: for example, we compared patients who completed full vaccination before first infection and received a post-infection Ad5-nCoV vaccine dose, with those who completed full vaccination before first infection and did not receive post-infection vaccination; this approach was repeated for inactivated vaccines and recombinant protein vaccines (**Table S5**). In this analysis, the types of vaccines administered before infection often aligned between the post-infection vaccination group and the post-infection non-vaccination group. Post-infection Ad5-nCoV vaccine dose given to those had received full vaccination before the first infection was associated with lower hazard of reinfection (0.67, 95% CI: 0.56, 0.80); a protective effect was also observed for other vaccine types: inactivated vaccines (0.92, 95% CI: 0.87, 0.98) or recombinant protein vaccines (0.77, 95% CI: 0.64, 0.92). We performed an additional ad-hoc sensitivity analysis using a Cox model that was not adjusted for the severity of the first infection. As shown in **Figure S2**, our findings were not affected by the non-inclusion of disease severity in the survival analysis model.

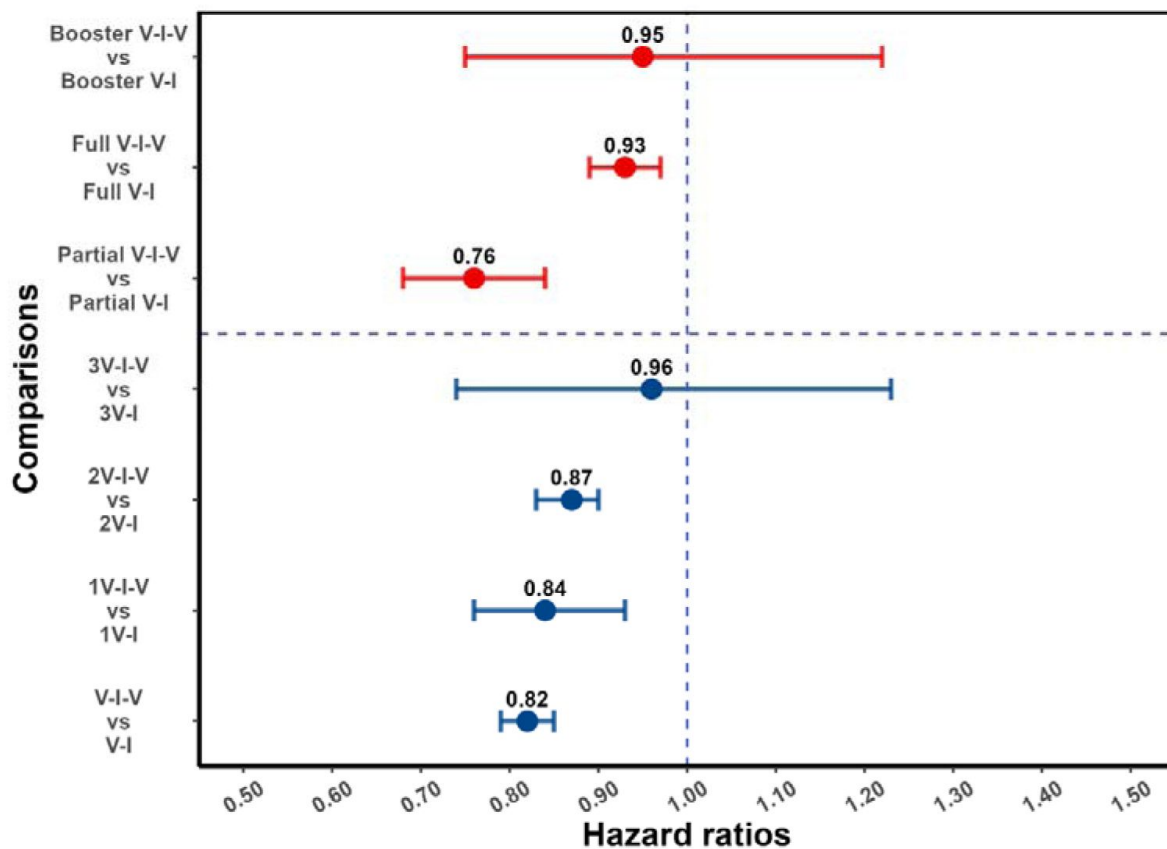


Figure 3.

Effect of post-infection vaccination on SARS-CoV-2 reinfection stratified by pre-infection vaccination.

Error bars (95% CIs) and circles represent aHR for SARS-CoV-2 reinfection estimated using Cox proportional hazards models. V-I-V, 1V-I-V, 2V-I-V, 3V-I-V corresponds to any pre-infection vaccination, 1, 2 and 3 vaccine doses before infection, then vaccination, respectively; they were compared to V-I, 1V-I, 2V-I, 3V-I, respectively. Partial V-I-V, Full V-I-V and Booster V-I-V represent partial vaccination, full vaccination and booster vaccination before infection, followed by post-infection vaccination, respectively. The number of doses received by individuals with partial versus full (and full with booster) vaccination depends on the type of SARS-CoV-2 vaccine received; in Table S3 we present a cross-classification of participants in the analytic population by these vaccination-related categorical variables.

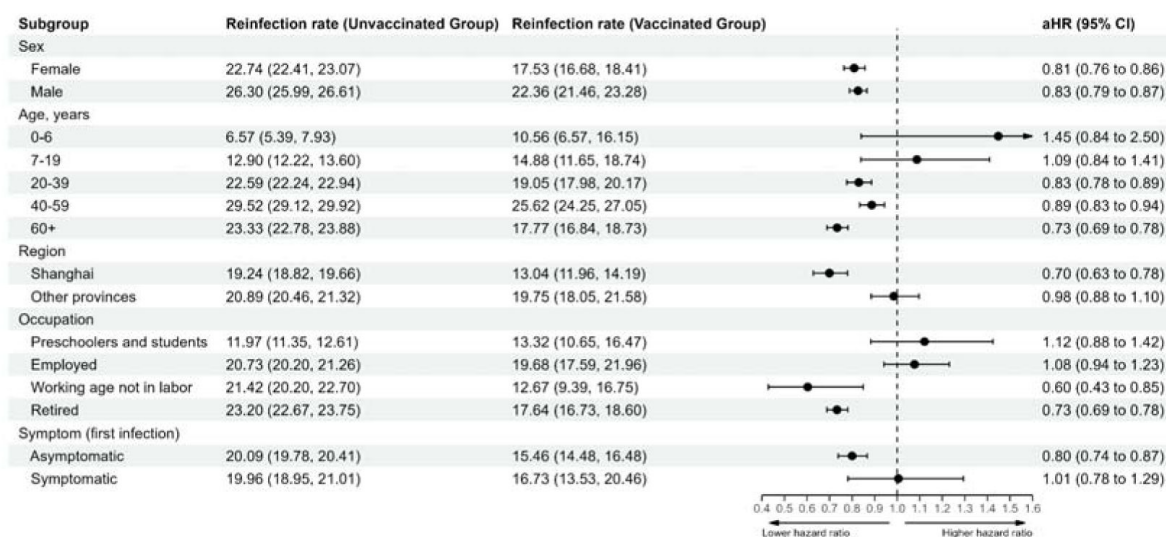


Figure 4.
Vaccine-related protection against SARS-CoV-2
reinfection stratified by demographic characteristic.

The vertical dotted line at 1.0 indicates no effect on protection.

3 Discussion

We aimed to estimate the added value of a vaccine dose given after SARS-CoV-2 infection in individuals who had received COVID-19 vaccination before the emergence of the Omicron variant. Although quantification of the protective effect of post-infection vaccination in individuals who have been both exposed to infection and previously vaccinated is a public health policy priority, as in many settings this group represents a large fraction of the population, studies on this question are often complicated by the variable timing of infection, vaccination and reinfection. Here, we leveraged the epidemiological history of COVID-19 in Shanghai municipality, where exposure to first and second SARS-CoV-2 infections occurred during well-defined periods of time, and large electronic databases allowed analyses of vaccination and infection history. Our study showed that post-infection vaccination reduced the risk of reinfection with the Omicron variant during a second, large surge of infections in the city. We also assessed the impact of the timing of vaccination relative to the second Omicron variant wave, and presented secondary and sensitivity analyses. This study will be used to guide COVID-19 vaccination policy in the municipality of Shanghai, and have implications for the rest of the country.

In Shanghai, COVID-19 vaccination started in February 2021 for individuals aged between 18 and 59 years, and over the following months, both older residents and children were included in the vaccination programme[9]. By the time of the first surge of Omicron variant infections, less than 70% of Shanghai residents aged 60 to 79 years had completed a primary vaccination series, and 40% had received a booster dose[8]. This was the context in which the population in this municipality was exposed to a first large epidemic of SARS-CoV-2 infections. The change in policy in December 2022 that was associated with a significant increase in the number of infections was our primary motivation to study the additional protection afforded by post-infection vaccination. Indeed, although epidemiological studies have been reported on the effect of vaccination against reinfection, primarily in Europe[10], North America[11] and Middle East[12], there is limited evidence from settings similar to that in Shanghai, where both the infection and vaccination histories, in terms of vaccine types, differ considerably from those in most Western countries. As individuals increasingly question the benefit of getting vaccinated after COVID-19 recovery, evidence generated locally is necessary.

Previous studies have shown that infection alone can result in immune responses that protect against Omicron variant reinfection. For example, a study in Qatar, with a test-negative design, found that an earlier Omicron variant infection provided protection against symptomatic BA.4 or BA.5 reinfection[13]. There are also different types of evidence that support post-infection vaccination. In Israel, post-infection vaccination protected against reinfection in individuals with no history of pre-infection vaccination[14], and prior SARS-CoV-2 infection was associated with lower risk for breakthrough infection among individuals receiving the BNT162b2 or mRNA-1273 vaccines in Qatar[15]. Further, a recent cohort study, performed in the US, of previously infected individuals who were unvaccinated at the time of first infection suggest that vaccination after recovery from COVID-19 decreased risk of reinfection by approximately half[16], which was consistent with a case-control study also conducted in the US[17]. Although our research question was different from those in these studies, data from Shanghai suggest that post-infection vaccination in individuals who had received at least one dose of SARS-CoV-2 vaccine before infection provides protection against reinfection. We note that whilst post-infection vaccination was protective for individuals who had received one or two vaccine doses before the first Omicron variant wave, for those individuals who had received three pre-infection vaccine doses, the evidence for a protective effect was limited. A possible explanation relates to the timing of post-infection vaccination: for all individuals with three pre-infection vaccine doses, the post-infection dose was given in December 2022 and, as a few weeks are required for immunologic boosting, the relatively short follow-up might have been insufficient to detect an effect. Similarly, evidence that

post-infection vaccination was protective for individuals who had received booster vaccination before the spread of the first Omicron variant wave in Shanghai was limited; the majority (793/810) of these individuals had 3 doses before first Omicron variant infection.

In a secondary analysis that assessed the impact of the timing of the post-infection vaccination on the estimated effect, we observe that vaccination within 30 and 90 days before the second surge of Omicron variant provided different degrees of protection, consistent with waning of immunity even in individuals with multiple vaccine doses and history of infection[18]. As the duration of follow-up in our study was relatively short, we were not able to analyze immunity duration over longer periods of time after the start of the second Omicron variant wave. This observation has implications for COVID-19 vaccination programmes: for example, under perfectly functioning logistics, it is possible that timing of vaccination campaigns, including for individuals who have been both previously infected and vaccinated, might be optimal if launched immediately after increase in incidence (e.g. linked to a new variant), rather than immediately after the end of a wave of infections.

In our analyses stratified by demographic characteristics, the protective effect of post-infection vaccination was higher among patients who were 60 years of age or older than among younger individuals. As older individuals are more likely to suffer severe disease, including in settings similar to Shanghai[6], this finding suggests that vaccination of individuals in this age group, even after previous infection and vaccination, is beneficial. Similar to the findings of Ghorbani et al[19], we also found that the rate of reinfection in females was lower than in males; however, the protective effect of post-infection vaccination was similar in the two groups.

We also observed greater protection after an Ad5-nCoV post-infection vaccination in individuals who were fully vaccinated before their first SARS-CoV-2 infection compared to post-infection vaccination with other vaccine types. Note however that not all post-infection doses were of the same type of pre-infection vaccine doses. Another stratified analysis revealed that the protective effect of additional doses was different in individuals who had asymptomatic presentations of the first infection versus those who had symptomatic disease (aHR 0.80 [0.74, 0.87] and 1.01 [0.78, 1.29]). This could be related to different immune responses after infections with different severities. Indeed, reduced antibody response after the initial SARS-CoV-2 infection has been associated with incidence of reinfections. For example, in the study by Islamoglu and colleagues[20], it was observed that antibody responses against SARS-CoV-2 were protective against COVID-19 reinfection. Consistently, in the Republic of Korea, CD4⁺ T-cell responses tended to be greater in patients who had severe disease[21]. In agreement with these studies, that had different designs, our analysis suggests that individuals who had severe or critical COVID-19 had a lower risk of reinfection compared to those who were asymptomatic (see **Table 1**). Note that individuals who were asymptomatic and those with mild symptoms during the first infection had comparable rates of reinfection. However, individuals with unknown disease severity during the first infection had a higher reinfection rate compared to those with documented clinical severity.

Strengths of our study include: the well-defined timing of first infections and reinfections in Shanghai, linked to the two Omicron variant waves; detailed information on vaccination based on a citywide system; mass screening for SARS-CoV-2 infections during the first Omicron variant surge; free testing during the second Omicron variant wave. Further, to prevent immortal time bias, a common problem in epidemiological studies where treatment/exposure assignment is not aligned with the start of follow-up, we performed survival analysis that used a time varying exposure variable. The insights from our study are also likely generalizable to similar urban settings facing distinct waves of SARS-CoV-2 infections and that adopted the same vaccine types. However, our study also has limitations. Firstly, information on key variables, such as occupation, household registration, clinical severity during the first infection, was missing for a non-negligible fraction of the study population. Although in the primary analysis we performed adjustment for likely confounders, and in the secondary analysis we used propensity score matching, as in other

observational studies, we cannot rule out the possibility of residual confounding, which in this context could be related to comorbidities that might affect both the decision to vaccinate after infection and risk of reinfection. Differences in healthcare-seeking behavior could also bias case ascertainment between post-infection vaccinated and unvaccinated individuals[22]. Although we restricted the study population to individuals who had received at least one pre-infection vaccination, which might have led to a higher degree of homogeneity in healthcare-seeking behaviour compared to that in the total population, it is possible that this bias might have affected our estimates. For example: individuals who were more health conscious might have been more likely to receive post-infection vaccination and also more likely to seek medical care or testing when reinfected, and this would have biased results toward the null; it is, however, also conceivable that these individuals were more likely to avoid contact with potentially infectious persons, which could have biased results in the opposite direction. Finally, data on the severity of infections during the second wave were not available, which prevented analyses of clinical outcomes other than infections (e.g. COVID-19-related hospitalization or death).

Although some previous studies [23–25] estimated similar or higher vaccine effectiveness against severe outcomes compared to outcomes that presumably include both milder and severe presentations, this pattern was not observed in all studies. Epidemiologists and public health officials who will use our results to define vaccination policy should thus take into account the fact that our analysis does not capture all benefits of post-infection vaccinations.

Conclusions

Although SARS-CoV-2 vaccination, including booster doses, is recommended by the public health authorities in Shanghai to reduce the local disease burden caused by COVID-19, there is increasing unwillingness in the population to receive additional vaccine doses, and vaccine fatigue has been frequently reported. Our study provides evidence that there is additional value for individuals who have been vaccinated in receiving vaccine doses after infections. It also suggests that vaccination programmes need to be linked to efficient surveillance for new infections so that public health authorities can maximize impact of additional doses, including in this group of patients.

4 Materials and methods

4.1 Study setting and participants

During the first Omicron variant wave, the entire city of Shanghai entered a lockdown phase on April 1, 2022; and on June 1, 2022, the local government declared the end of the city-wide lockdown[6]. Residents (citizens, including immigrants from other provinces, and foreigners) in Shanghai, a provincial-level municipality in China with a population of more than 25 million people, underwent several rounds of SARS-CoV-2 RT-PCR testing between April 1 and May 31, 2022. This study included individuals diagnosed with their first SARS-CoV-2 infection between April 1 and May 31, 2022 in the Pudong District, which is a large and densely populated district of Shanghai spanning an area of 1,210 square kilometers with a permanent resident population of 5.57 million, served by more than 30 hospitals and 60 community health centers[26]; both individuals who were diagnosed by mass screening and those with symptoms who were seen by healthcare professionals were included. Information on infection history as well as data on demographic variables (sex at birth, and age) were provided by the Center for Disease Control and Prevention in Shanghai, China.

Additional information (e.g. on occupation, residence, clinical severity and symptoms of first infection) was available for patients with hospital records and those who were transferred to a hospital. The recorded clinical severity was categorized as asymptomatic, mild, moderate, severe, or critically ill[27 [↗](#), 28 [↗](#)]. During the first Omicron variant wave in Shanghai, many Fangcang shelter hospitals were rapidly converted into facilities to treat COVID-19 patients and made important contributions in providing adequate healthcare to patients with mild to moderate symptoms, and preventing further viral transmission in the community[29 [↗](#)]. The efficient referral and transfer mechanisms in the local communities meant that the majority of patients were admitted to Fangcang shelter hospitals one or two days after testing positive for SARS-CoV-2[30 [↗](#)]. However, many patients were either admitted to hospital without complete clinical information or not transferred to other hospitals; for these study participants, information on clinical severity was often missing.

4.2 Study design and eligibility criteria

There was a second surge of Omicron variant cases from December 2022; and from January 2023, free nucleic acid testing services were no longer offered in Shanghai, and mandatory PCR testing on all personnel ceased. After this change, the cost of an individual test was 16 yuan (US\$2.33) [31 [↗](#)]. For this reason, the outcome variable in our analysis was based on reinfection data collected prior to January 2023. Reinfection-related death was defined as death within 30 days of a SARS-CoV-2 reinfection[10 [↗](#)]; individuals who died from reasons unrelated to COVID-19 between the two Omicron variant waves were excluded. Individuals were also excluded if the date of first infection was missing. As the reasons why some individuals refused to receive vaccination varied and were often unknown[32 [↗](#)], our analysis focused on individuals who had received at least one vaccine dose before the first SARS-CoV-2 infection.

4.3 Vaccination and reinfection data

The Shanghai Group Immunization System captures all vaccine administrations in the municipality and is updated daily. This system is linked to the National Immunization Program Information System, which also includes national identification-matched COVID-19 vaccinations received outside of Shanghai[8 [↗](#)]. Vaccination status was categorized in accordance with national technical recommendations for COVID-19 vaccination[8 [↗](#)]: unvaccinated, partially vaccinated, fully vaccinated and fully vaccinated with booster dose (here, also referred to as booster vaccination). Here, inactivated vaccines refer to Sinovac-CoronaVac, Sinopharm/BIBP COVID-19 vaccine, and Sinopharm/WIBP COVID-19 vaccine; Ad5-vectored vaccine refers to Cansino Ad5-nCoV-S COVID-19 vaccine; and recombinant protein vaccine refers to recombinant COVID-19 vaccine (CHO cell), Anhui Zhifei Longcom Biopharmaceutical Institute of Microbiology.

As in other epidemiological studies[19 [↗](#), 33 [↗](#)], we defined reinfection as a positive SARS-CoV-2 RT-PCR or rapid antigen test at least 90 days after the first positive test. Phylogenetic analysis coupled with contact tracing data revealed community transmission of Omicron BA.5.2 sublineage in Shanghai[9 [↗](#)]; the subvariant was estimated to have caused ~90% of infections during the second Omicron variant wave.

4.4 Statistical analysis

Continuous variables are summarized as medians and interquartile ranges (IQR), and categorical variables, as counts, proportions or percentages. Cumulative incidence of reinfections was calculated and expressed as the number of reinfected individuals per 100 participants. Cox proportional hazards models with a time varying exposure variable corresponding to post-infection vaccination were used to estimate adjusted hazard ratios (aHR). In this analysis, time to reinfection was the outcome, and post-infection vaccination, the exposure of interest; models were adjusted for sex, age, residence, occupation and clinical severity of the first SARS-CoV-2 infection. As, due to social distancing measures, the number of SARS-CoV-2 infections diagnosed between

May and November 2022 in Shanghai was low ($N = 89$ individuals), the start of the follow-up in the survival analysis was on December 1, and for each participant, the follow-up continued until January 3, 2023 or confirmed SARS-CoV-2 reinfection, whichever occurred earlier. We note that by using this approach, both the start of the follow-up and the eligibility (which required confirmed SARS-CoV-2 infection during the initial Omicron variant wave, and being at risk of infection when the second Omicron variant wave occurred in December) are temporally aligned. On the other hand, for some participants, the time of exposure (post-infection vaccination) occurred after time zero, which could potentially lead to immortal time bias[34]; for this reason, the exposure variable was as time-varying. We also assessed whether this effect was modified by demographic characteristics.

We conducted a secondary analysis to estimate the protection afforded by post-infection vaccination before the second Omicron variant wave (that is, only vaccine doses received before December 2022 were used in defining the exposure variable); this analysis was stratified by time intervals between post-infection vaccination and the second Omicron variant wave, and by post-infection vaccine type (for individuals who had the same vaccination status or the same number of doses before first infection). For this secondary analysis, propensity score matching was used to improve comparability between the exposed and unexposed groups; the propensity score was calculated using a logistic regression model with all available baseline characteristics, and a one-to-one matching was performed using the nearest neighbor matching method with a caliper width of 0.20[35]. We assessed the balance of covariates after matching using standardized mean differences (SMD), and considered a value of less than 0.1 to be indicative of adequate matching (Table S6-8). All statistical analyses were performed using R.4.1.1 software (Foundation for Statistical Computing, Vienna, Austria; <https://www.r-project.org>). We followed the Strengthening the reporting of observational studies in epidemiology (STROBE) recommendations, and the STROBE checklist is provided in the *Supplementary Appendix* (Table S9)

Acknowledgements

We gratefully acknowledge all participants in this study.

Additional information

Funding

Funder	Grant reference number	Author
Key Discipline Program of Pudong New Area Health System	PWZxk2022-25	Pengfei Deng
Development and Application of Intelligent Epidemic Surveillance and Artificial Intelligence Analysis System	21002411400	Weibing Wang
Shanghai Three-year Action Plan to Strengthen the Construction of Public Health System	GWVI-11.2-XD08	Caoyi Xue
Key Disciplines supported by Shanghai Municipal Health Commission	GWVI-11.1-02	Caoyi Xue
Shanghai Municipal Health Commission Clinical Research Program	20214Y0020	Ye Yao
General Program of Natural Science Foundation of Shanghai Municipality	22ZR1414600	Ye Yao
Young Health Talents Program of Shanghai Municipality	2022YQ076	Ye Yao
The funders had no role in study design, data collection and interpretation, or the decision to submit the work for publication.		

Author contributions

BZ, BG, WW and YY designed the study. CX, PD and BZ extracted the data and constructed the database. BZ, JT and XL analyzed the data. BZ and WW drafted the manuscript—original draft. BG, XC, YY and WW conducted critical revisions of the manuscript—review and editing. All authors read and approved the final manuscript. **Ethics**

The study was approved by the Ethical Review Committee in the Shanghai Pudong New Area Center for Disease Control and Prevention (approval number: PDCDCLL-20220801-001).

Consent for publication

Not applicable.

Competing interests

The authors declare that they have no known competing financial interest or personal relationship that could influence the work reported in this paper.

Additional files

Supplementary files

Table S1. Descriptive overview of the study population.

Table S2. Characteristics of vaccination distribution among study population.

Table S3. Cross-classification based on the number of vaccine doses received before the first infection and vaccination status.

Table S4. Effect of post-infection vaccination in individuals with no history of vaccination before infection.

Table S5. Effect of vaccination after first infection against SARS-CoV-2 reinfection stratified by sequence of vaccination dose and vaccination status before first infection. Table S6. Baseline demographic characteristics of unvaccinated and vaccinated groups after first infection stratified by dose received before first infection.

Table S7. Baseline demographic characteristics of unvaccinated and vaccinated groups after first infection stratified by intervals before the policy change.

Table S8. Baseline demographic characteristics of unvaccinated and vaccinated groups after first infection stratified by vaccination status prior infection.

Table S9. STROBE Statement.

Figure S1. Vaccination status coverage in the study population.

Figure S2. Effect of post-infection vaccination on SARS-CoV-2 reinfection stratified by pre-infection vaccination and not adjusted for the severity of the first infection.

Data availability

The individual-level data used in this study are sensitive and cannot be publicly shared. Requests for data should be made to the Shanghai Pudong New Area Center for Disease Control and Prevention.

References

1. Pan Y, Wang L, Feng Z, Xu H, Li F, Shen Y, Zhang D, Liu WJ, Gao GF, Wang Q (2023) **Characterisation of SARS-CoV-2 variants in Beijing during 2022: an epidemiological and phylogenetic analysis** *Lancet* **401**:664–672
2. Zhu N *et al.* (2020) **A Novel Coronavirus from Patients with Pneumonia in China, 2019** *N Engl J Med* **382**:727–733
3. Viana R *et al.* (2022) **Rapid epidemic expansion of the SARS-CoV-2 Omicron variant in southern Africa** *Nature* **603**
4. Zhou Y, Zhi H, Teng Y (2023) **The outbreak of SARS-CoV-2 Omicron lineages, immune escape, and vaccine effectivity** *J Med Virol* **95**
5. Li J, Lai S, Gao GF, Shi W (2021) **The emergence, genomic diversity and global spread of SARS-CoV-2** *Nature* **600**
6. Chen X *et al.* (2022) **Estimation of disease burden and clinical severity of COVID-19 caused by Omicron BA.2 in Shanghai, February-June 2022** *Emerg Microbes Infect* **11**:2800–2807
7. Zheng L, Liu S, Lu F (2023) **Impact of National Omicron Outbreak at the end of 2022 on the future outlook of COVID-19 in China** *Emerg Microbes Infect* **12**
8. Huang Z *et al.* (2023) **Effectiveness of inactivated COVID-19 vaccines among older adults in Shanghai: retrospective cohort study** *Nat Commun* **14**
9. Huang Z *et al.* (2022) **Effectiveness of inactivated and Ad5-nCoV COVID-19 vaccines against SARS-CoV-2 Omicron BA. 2 variant infection, severe illness, and death** *BMC Med* **20**
10. Nielsen KF *et al.* (2022) **Vaccine effectiveness against SARS-CoV-2 reinfection during periods of Alpha, Delta, or Omicron dominance: A Danish nationwide study** *PLoS Med* **19**
11. Bobrovitz N *et al.* (2023) **Protective effectiveness of previous SARS-CoV-2 infection and hybrid immunity against the omicron variant and severe disease: a systematic review and meta-regression** *Lancet Infect Dis* **23**:556–567
12. Gazit S, Saciuk Y, Perez G, Peretz A, Ben-Tov A, Stuart EA, Patalon T (2023) **Hybrid immunity against reinfection with SARS-CoV-2 following a previous SARS-CoV-2 infection and single dose of the BNT162b2 vaccine in children and adolescents: a target trial emulation** *Lancet Microbe* **4**:e495–e505
13. Altarawneh HN *et al.* (2022) **Protective Effect of Previous SARS-CoV-2 Infection against Omicron BA.4 and BA.5 Subvariants** *N Engl J Med* **387**:1620–1622
14. Hammerman A, Sergienko R, Friger M, Beckenstein T, Peretz A, Netzer D, Yaron S, Arbel R (2022) **Effectiveness of the BNT162b2 Vaccine after Recovery from Covid-19** *N Engl J Med* **386**:1221–1229

15. Abu-Raddad LJ *et al.* (2021) **Association of Prior SARS-CoV-2 Infection With Risk of Breakthrough Infection Following mRNA Vaccination in Qatar** *Jama* **326**:1930–1939
16. Lewis N, Chambers LC, Chu HT, Fortnam T, De Vito R, Gargano LM, Chan PA, McDonald J, Hogan JW (2022) **Effectiveness Associated With Vaccination After COVID-19 Recovery in Preventing Reinfection** *JAMA Netw Open* **5**
17. Cavanaugh AM, Spicer KB, Thoroughman D, Glick C, Winter K (2021) **Reduced Risk of Reinfection with SARS-CoV-2 After COVID-19 Vaccination - Kentucky, May-June 2021** *MMWR Morb Mortal Wkly Rep* **70**:1081–1083
18. Ferdinands JM *et al.* (2022) **Waning of vaccine effectiveness against moderate and severe covid-19 among adults in the US from the VISION network: test negative, case-control study** *Bmj* **379**
19. Sotoodeh Ghorbani S, Taherpour N, Bayat S, Ghajari H, Mohseni P, Hashemi Nazari SS (2022) **Epidemiologic characteristics of cases with reinfection, recurrence, and hospital readmission due to COVID-19: A systematic review and meta-analysis** *J Med Virol* **94**:44–53
20. Islamoglu MS *et al.* (2021) **Relationship between Antibody Levels and SARS-Cov-2 Reinfection** *Ann Clin Lab Sci* **51**:750–755
21. Kang CK *et al.* (2021) **Longitudinal Analysis of Human Memory T-Cell Response According to the Severity of Illness up to 8 Months After Severe Acute Respiratory Syndrome Coronavirus 2 Infection** *J Infect Dis* **224**:39–48
22. Yung CF, Pang D, Kam KQ, Lye DC, Ong B, Chong CY, Tan KB (2023) **BNT162b2 vaccine protection against omicron and effect of previous infection variant and vaccination sequence among children and adolescents in Singapore: a population-based cohort study** *Lancet Child Adolesc Health*
23. Magen O, Waxman JG, Makov-Assif M, Vered R, Dicker D, Hernán MA, Lipsitch M, Reis BY, Balicer RD, Dagan N (2022) **Fourth Dose of BNT162b2 mRNA Covid-19 Vaccine in a Nationwide Setting** *N Engl J Med* **386**:1603–1614
24. Nasreen S *et al.* (2022) **Effectiveness of COVID-19 vaccines against symptomatic SARS-CoV-2 infection and severe outcomes with variants of concern in Ontario** *Nat Microbiol* **7**:379–385
25. Sacco C *et al.* (2022) **Effectiveness of BNT162b2 vaccine against SARS-CoV-2 infection and severe COVID-19 in children aged 5-11 years in Italy: a retrospective analysis of January-April, 2022** *Lancet* **400**:97–103
26. **Pudong New Area Builds Full-Coverage Medical Service Chain** [<https://enshanghaiacademy.shu.edu.cn/info/1013/2339.htm>] no date
27. Chen Z *et al.* (2022) **Epidemiological characteristics and transmission dynamics of the outbreak caused by the SARS-CoV-2 Omicron variant in Shanghai, China: A descriptive study** *Lancet Reg Health West Pac* **29**
28. Ge Y *et al.* (2021) **COVID-19 Transmission Dynamics Among Close Contacts of Index Patients With COVID-19: A Population-Based Cohort Study in Zhejiang Province, China** *JAMA Intern Med* **181**:1343–1350

29. Zhang X, Zhang W, Chen S (2022) **Shanghai's life-saving efforts against the current omicron wave of the COVID-19 pandemic** *Lancet* **399**:2011–2012
30. Ye L, Li WF, Shao J, Xu Z, Ju J, Xu H (2022) **Fighting Omicron epidemic in China: Real-world big data from Fangcang shelter hospital during the outbreak in Shanghai 2022** *J Infect* **85**:436–480
31. **No more free PCR tests in Shanghai from Sunday** [<https://www.shine.cn/covid19/2301064857/>] no date
32. Wu J, Li Q, Silver Tarimo C, Wang M, Gu J, Wei W, Ma M, Zhao L, Mu Z, Miao Y (2021) **COVID-19 Vaccine Hesitancy Among Chinese Population: A Large-Scale National Study** *Front Immunol* **12**
33. **Investigative criteria for suspected cases of SARS-CoV-2 reinfection (ICR)** [<https://stacks.cdc.gov/view/cdc/96072>] no date
34. Hernán MA, Sauer BC, Hernández-Díaz S, Platt R, Shrier I (2016) **Specifying a target trial prevents immortal time bias and other self-inflicted injuries in observational analyses** *J Clin Epidemiol* **79**:70–75
35. Ueno M *et al.* (2021) **Prognostic impact of neoadjuvant chemotherapy in patients with synchronous colorectal liver metastasis: A propensity score matching comparative study** *Int J Surg* **94**

Editors

Reviewing Editor

Joshua Schiffer

Fred Hutchinson Cancer Research Center, Seattle, United States of America

Senior Editor

Eduardo Franco

McGill University, Montreal, Canada

Reviewer #1 (Public Review):

Summary:

Zheng and colleagues assessed the real world efficacy of SARS-CoV-2 vaccination against re-infection following the large omicron wave in Shanghai in April, 2022. The study was performed among previously vaccinated individuals. The study successfully documents a small but real added protective benefit of re-vaccination, though this diminishes in previously boosted individuals. Unsurprisingly, vaccine preventative efficacy was higher if the vaccine was given in the month before the 2nd large wave in Shanghai. The re-infection rate of 24% suggests that long-term anti-COVID immunity is very difficult to achieve. The conclusions are largely supported by the analyses. These results may be useful for planning the timing of subsequent vaccine rollouts.

Strengths:

The strengths of the study are a very large and unique cohort based on synchronously timed single infection among individuals with well documented vaccine histories. Statistical

analyses seem appropriate. As with any cohort study, there are potential confounders and the possibility of misclassification and the authors outline limitations nicely in the discussion.

Weaknesses:

The authors have addressed each of my points thoroughly.

<https://doi.org/10.7554/eLife.94990.2.sa2>

Reviewer #2 (Public Review):

Summary:

This paper evaluates the effect of COVID-19 booster vaccination on reinfection in Shanghai, China among individuals who received primary COVID-19 vaccination followed by initial infection, during an Omicron wave.

Strengths:

A large database is collated from electronic vaccination and infection records. Nearly 200,000 individuals are included in the analysis and 24% became reinfected.

Weaknesses:

The authors have revised the manuscript and have provided satisfactory responses to my prior comments.

<https://doi.org/10.7554/eLife.94990.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public Review):

Summary:

Zheng and colleagues assessed the real-world efficacy of SARS-CoV-2 vaccination against re-infection following the large omicron wave in Shanghai in April 2022. The study was performed among previously vaccinated individuals. The study successfully documents a small but real added protective benefit of re-vaccination, though this diminishes in previously boosted individuals. Unsurprisingly, vaccine preventative efficacy was higher if the vaccine was given in the month before the 2nd large wave in Shanghai. The re-infection rate of 24% suggests that long-term anti-COVID immunity is very difficult to achieve. The conclusions are largely supported by the analyses. These results may be useful for planning the timing of subsequent vaccine rollouts.

Strengths:

The strengths of the study are a very large and unique cohort based on synchronously timed single infection among individuals with well-documented vaccine histories. Statistical analyses seem appropriate. As with any cohort study, there are potential confounders and the possibility of misclassification and the authors outline limitations nicely in the discussion.

Weaknesses:

(1) Partially and fully vaccinated are never defined and it is difficult to understand how this differs from single, and double, booster vaccines. The figures including all of these groups are a bit confusing for this reason.

We agree with the reviewer that the distinction between these groups could have been made clearer. To address this comment, we modified the legend of the figure that presents hazard ratios based on these two categorisations (here, and throughout this document, changes in the text are underlined):

“Figure 3. Effect of post-infection vaccination on SARS-CoV-2 reinfection stratified by pre-infection vaccination. Error bars (95% CIs) and circles represent aHR for SARS-CoV-2 reinfection estimated using Cox proportional hazards models. V-I-V, 1V-I-V, 2V-I-V, 3V-I-V corresponds to any pre-infection vaccination, 1, 2 and 3 vaccine doses before infection, then vaccination, respectively; they were compared to V-I, 1V-I, 2V-I, 3V-I, respectively. Partial V-I-V, Full V-I-V and Booster V-I-V represent partial vaccination, full vaccination and booster vaccination before infection, followed by post-infection vaccination, respectively. The number of doses received by individuals with partial versus full (and full with booster) vaccination depends on the type of SARS-CoV-2 vaccine received; in Table S3 we present a cross-classification of participants in the analytic population by these vaccination-related categorical variables.”

Further, to facilitate visualisation of Figure 3, and emphasize that estimates are presented based on two different ways of categorising vaccination history, we have now included a horizontal line between estimates based on each category.

Table S3 has been included in the Supplementary Appendix:

(2) Figure 3 is a bit challenging to interpret because it is a bit atypical to compare each group to a different baseline (ie 2V-I-V vs 2V-I). I would label the y-axis 2V-I-V vs 2V-I (change all of the labels) to make this easier to understand.

We agree that having the y-axis tick labels describing both groups being compared, rather than only describing the post-infection vaccination group, will help readers to understand this figure. In our response to the previous comment, we presented an updated version of this figure, where this change was also incorporated (see above).

(3) A 15% reduction in infection is quite low. It would be helpful to discuss if any quantitative or qualitative signals suggest at least a reduction in severe outcomes such as death, hospitalization, ER visits, or long COVID. I am not sure that a 15% reduction in cases supports extra vaccination without some other evidence of added benefit.

Unfortunately, data on the clinical severity of diagnosed SARS-CoV-2 infections were not available. Some previous studies on COVID-19 vaccines observed that effectiveness against severe outcomes was similar or higher than that for outcomes that do not imply severe disease (e.g. infection). For example, in a study in Israel comparing four versus three vaccine doses, Magen and colleagues observed that the effectiveness of a fourth dose, relative to three doses, was 52% against infection, 61% against symptomatic COVID-19, and 76% against COVID-19 related death (Magen et al. *Fourth Dose of BNT162b2 mRNA Covid-19 Vaccine in a Nationwide Setting*. NEJM 2022; see also, for example, Nasreen et al. *Effectiveness of COVID-19 vaccines against symptomatic SARS-CoV-2 infection and severe outcomes with variants of concern in Ontario*. Nature Microbiology 2022, or Sacco et al. *Effectiveness of BNT162b2 vaccine against SARS-CoV-2 infection and severe COVID-19 in children aged 5–11 years in Italy: a retrospective analysis of January–April, 2022*. Lancet 2022). However, this pattern of

increasing effectiveness with increasing outcome severity was not consistently reported in all studies or settings. We agree that public health officials who will use our results to guide future vaccination policy in China and abroad need to interpret the results in the context of these other outcomes that were not assessed and of those previous studies, that, although performed in different epidemiological settings, suggest that our analysis does not capture all benefits of post-infection vaccine doses.

We have now included the following statements in the *Discussion* section:

“Finally, data on the severity of infections during the second wave were not available, which prevented analyses of clinical outcomes other than infections (e.g. COVID-19-related hospitalization or death). Although some previous studies (Magen et al. Fourth Dose of BNT162b2 mRNA Covid-19 Vaccine in a Nationwide Setting. *NEJM* 2022; Nasreen et al. Effectiveness of COVID-19 vaccines against symptomatic SARS-CoV-2 infection and severe outcomes with variants of concern in Ontario. *Nature Microbiology* 2022) estimated similar or higher vaccine effectiveness against severe outcomes compared to outcomes that presumably include both milder and severe presentations, this pattern was not observed in all studies. Epidemiologists and public health officials who will use our results to define vaccination policy should thus take into account the fact that our analysis does not capture all benefits of post-infection vaccinations.”

(4) Why exclude the 74962 unvaccinated from the analysis. it would be interesting to see if getting vaccinated post-infection provides benefits to this group

The reasons why we focused on individuals who had been vaccinated before their first infection were two: (i) in most settings, including those with SARS-CoV-2 epidemiologic history similar to that of Shanghai, a high percentage of the population has received vaccine doses; (ii) in settings with high vaccination coverage, the group of individuals who remain unvaccinated despite widespread availability of vaccines likely differs from those who have been vaccinated – for example, with regard to behavioural factors and comorbidity profile. Having said that, we agree that reporting analyses for the group of individuals who had not been vaccinated before first infection might be informative. We have thus included in the Supplementary Appendix a short section that reports results for this group of patients; Table S4 also presents these estimates.

“Effect of post-infection vaccination in individuals with no history of vaccination before infection

In this supplementary section, we present findings for individuals who were unvaccinated before infection during the first Omicron variant wave in Shanghai. For this group of individuals, post-infection vaccination did not confer significant protection against reinfection (adjusted hazard ratio [aHR] 1.06, 95% CI 0.97, 1.16). The analysis indicates that the effect of post-infection vaccine doses was not significant in both female (aHR 0.97 [0.84, 1.11]) and male individuals (aHR 1.12 [0.99, 1.26]), as well as for participants aged 60 years or older (aHR 0.92 [0.82, 1.04]) and younger adults (20–60 years) (aHR 1.12 [0.92, 1.37]). These results suggest that, in the context of the two Omicron variant waves in Shanghai, a first vaccine dose administered after infection did not provide a clear benefit in terms of reducing risk of subsequent infections for those not previously vaccinated.”

We refer to this new analysis in the Results section:

“For individuals who had received at least one vaccine dose before infection during the first Omicron variant wave, post-infection vaccination was protective against reinfection (adjusted hazard ratio [aHR] 0.82, 95% CI 0.79, 0.85). As shown in Figure 3, this protective effect was observed in subgroups defined by the number of pre-infection vaccine doses: aHR of 0.84 (95% CI, 0.76, 0.93) and 0.87 (95% CI, 0.83, 0.90) for one and two pre-infection doses

respectively; and for patients with three vaccine doses prior to infection, the association was not statistically significant (aHR: 0.96 [0.74, 1.23]). When analyses are stratified by partial and full vaccination status before the first infection, an additional vaccine dose was protective (aHR 0.76 [0.68, 0.84], and 0.93 [0.89, 0.97], respectively); and among individuals who had received booster vaccination before the spread of the first Omicron variant wave in Shanghai, the hazard ratio estimate was consistent with a more limited effect (aHR: 0.95 [0.75, 1.22]). For comparison, results for individuals who had not been vaccinated before their first infection are shown in the Supplementary Appendix (supplementary section “Effect of post-infection vaccination in individuals with no history of vaccination before infection” and Table S4)”

(5) *Pudong should be defined for those who do not live in China.*

We have now included a sentence defining Pudong in the *Methods* section:

“This study included individuals diagnosed with their first SARS-CoV-2 infection between April 1 and May 31, 2022 in the Pudong District, which is a large and densely populated district of Shanghai spanning an area of 1,210 square kilometers with a permanent resident population of 5.57 million, served by more than 30 hospitals and 60 community health centers;...”

(6) *The discussion about healthcare utilization bias is welcomed and well done. It would be great to speculate on whether this bias might favor the null or alternative hypothesis.*

We believe the reviewer is referring to the following statement:

“Differences in healthcare-seeking behavior could also bias case ascertainment between post-infection vaccinated and unvaccinated individuals, although, as we restricted the study population to individuals who had received at least one pre-infection dose, this potential bias might be more limited than in other vaccine studies.”

Bias linked to healthcare seeking behaviour could affect the association between vaccination and infection in two different ways: individuals who are more health conscious are more likely to get vaccinated and also to seek medical care when infected, and this would bias results toward null; however, if the same individuals are also more likely to avoid exposure to potentially infectious individuals, their behaviour could also bias results in the opposite direction – that is, it would appear to increase vaccine effectiveness. As mentioned in the Discussion section, we expected this bias to be limited. We have now modified the paragraph:

“Differences in healthcare-seeking behavior could also bias case ascertainment between post-infection vaccinated and unvaccinated individuals. Although we restricted the study population to individuals who had received at least one pre-infection vaccination, which suggests a higher degree of homogeneity in healthcare-seeking behaviour compared to that in the total population, it is possible that this bias might have affected our estimates. For example: individuals who were more health conscious might have been more likely to receive post-infection vaccination and also more likely to seek medical care or testing when reinfected, and this would have biased results toward the null; it is, however, also conceivable that these individuals were more likely to avoid contact with potentially infectious persons, which could have biased results in the opposite direction.”

Reviewer #2 (Public Review):

Summary:

This paper evaluates the effect of COVID-19 booster vaccination on reinfection in Shanghai, China among individuals who received primary COVID-19 vaccination followed

by initial infection, during an Omicron wave.

Strengths:

A large database is collated from electronic vaccination and infection records. Nearly 200,000 individuals are included in the analysis and 24% became reinfected.

Weaknesses:

The article is difficult to follow in terms of the objectives and individuals included in various analyses. There appear to be important gaps in the analysis. The electronic data are limited in their ability to draw causal conclusions.

More detailed comments:

In multiple places (abstract, introduction), the authors frame the work in terms of understanding the benefit of booster vaccination among individuals with hybrid immunity (vaccination + infection). However, their analysis population does not completely align with this framing. As best as I can tell, only individuals who first received COVID-19 vaccination, and subsequently experienced infection, were included. Why the analysis does not also consider individuals who were infected and then vaccinated is not clear.

The focus of our analysis is on the most frequent scenario in many countries: settings where a high proportion of the population has been vaccinated. As mentioned in our response to a comment from Reviewer #1, those individuals who remain unvaccinated after the first years of this pandemic are likely to be different, with respect to many factors, from individuals with history of SARS-CoV-2 vaccination. Further, differences between unvaccinated and vaccinated individuals are likely setting-specific, linked to local availability of and access to vaccination, cultural differences in healthcare seeking behaviour, and possible differences in the frequencies of medical conditions that might influence (promote or prevent) vaccine uptake. We prefer to keep the focus of this work on individuals who had been vaccinated before their first infection; however, we have now included in the Supplementary Appendix a section, presented in a response to Reviewer #1, that reports results for this group of individuals.

In vaccine effectiveness analyses, why was time since initial infection not examined as a modifier of the booster effect? Time since the onset of the Omicron wave is only loosely tied to the immune status of the individual.

We agree with the reviewer that assessing effect modification by the time since initial infection would be important. However, in Shanghai, most initial infections occurred during a narrow time window relative to the time window between the first and second Omicron variant waves. Indeed, as mentioned in the *Results* section, most first infections (243,906, 88.8%) occurred in April; for 306 (0.1%) individuals, information on the date of first infection was not available. Given this narrow time window and in order to limit the number of comparisons in our study, we preferred not to investigate this aspect of the hybrid immunity. In settings where multiple SARS-CoV-2 waves occurred, over a longer period of time, which would imply sufficient variation in this variable “time since initial infection”, we believe that it would be essential to account for this.

The effect of booster vaccination on preventing symptomatic vs. asymptomatic reinfection does not appear to have been evaluated; this is a key gap in the analysis and it would seem the data would support it.

Not having clinical presentation data is a limitation in our study. That is a weakness of many real-world vaccine effectiveness analyses based large medical and administrative datasets. We have now explicitly mentioned this in the Discussion section.

“Finally, data on the severity of infections during the second wave were not available, which prevented analyses of clinical outcomes other than infections (e.g. COVID-19-related hospitalization or death). Although some previous studies (Magen et al. Fourth Dose of BNT162b2 mRNA Covid-19 Vaccine in a Nationwide Setting. NEJM 2022; Nasreen et al. Effectiveness of COVID-19 vaccines against symptomatic SARS-CoV-2 infection and severe outcomes with variants of concern in Ontario. Nature Microbiology 2022) estimated similar or higher vaccine effectiveness against severe outcomes compared to outcomes that presumably include both milder and severe presentations, this pattern was not observed in all studies. Epidemiologists and public health officials who will use our results to define vaccination policy should thus take into account the fact that our analysis does not capture all benefits of post-infection vaccinations.”

In lines 105-108, the demographic description of the analysis population is incomplete. Is sex or gender identity being described? Are any individuals non-binary? What is the age distribution? (Only the proportions 20-39 and under 6 are stated.)

We have now clarified in the manuscript that only information on sex at birth was provided by the Center for Disease Control and Prevention in Shanghai. We made the following change in the Methods section:

“Information on infection history as well as data on demographic variables (sex at birth, and age) were provided by Center for Disease Control and Prevention in Shanghai, China”

We have also modified the legend of Table 1:

“Table 1. Characteristics of the study population and reinfection rate by post-infection vaccination status. Here, reinfection rate refers to the percentage of the relevant study subpopulation with evidence of reinfection between December 1, 2022 and January 3, 2023. Note that for the variables on region, occupation, and clinical severity, data are missing for large fractions of the study population. Note also that information was only available on sex at birth, but not on gender.”

Regarding the reviewer’s comment on the age distribution, this information is presented for the following categories in Table 1: 0-6 years, 7-19 years, 20-39 years, 40-59 years, and 60+ years. However, we had not referred to Table 1 in the section 3.1 of the manuscript. We have now corrected that:

“To assess the effect of an additional vaccine dose given after infection, the analytic sample consisted of 199,312 individuals (Figure 1). 85,804 were women (43.1%); 836 (0.4%) had gender information missing. 38.1% of the study participants were aged 20 to 39 years and only 0.9% were aged 0 to 6 years (see Table 1 for additional information).”

Figure 1 consort diagram is confusing. In the last row, are the two boxes independent or overlapping sets of individuals? Are all included in secondary analyses?

We agree that additional information should have been provided in the legend. The boxes represent overlapping sets of individuals – that is, some individuals were included in both secondary analyses in the box on the left and in the box on the right. These analyses involved different ways of categorizing individuals. Below is the updated figure legend:

“Figure 1. Flow chart describing the selection of participants for the analysis. The number of individuals in this figure is not the same as some of the numbers in Table 1 because of

missing data in key variables. Note that in the bottom part of the chart, related to secondary analyses, the boxes represent overlapping sets of study participants; in other words, some individuals included in the secondary analyses that correspond to the left box were also included in analyses corresponding to the box on the right.”

Recommendations for the authors:

Reviewer #2 (Recommendations For The Authors):

Minor comment: the terms "vaccination"/"vaccinated" are used both to refer to the primary vaccination (pre-initial infection) and to the booster vaccination (post-initial vaccination), and this causes confusion.

Thank you. We have now revised the manuscript (Methods, Results and Discussion sections) to use the terms “post-infection vaccination” and “post-infection vaccinated” to reduce ambiguity. We also included the following statement in the Background section:

“In December 2022, an important change in the COVID-19 policy in China, namely the end of most social distancing measures and of mass screening activities, was associated with a second surge in SARS-CoV-2 infections in Shanghai. The current circulation of the virus in the Shanghainese population and reports of vaccine fatigue mean that it is important to estimate the protective effect of vaccination against reinfection in this population. In this study, we aimed to quantify the effect of vaccine doses given after a first infection on the risk of subsequent infection. For that, we used data collected during the first Omicron variant wave, when hundreds of thousands of individuals tested real-time polymerase chain reaction (RT-PCR)-positive for SARS-CoV-2 infection⁸ in Shanghai, of which 275,896 individuals in Pudong. The fact that the population in Shanghai was mostly SARS-CoV-2 infection naïve before the spread of the Omicron variant provides a unique opportunity to estimate the real-world benefit of post-infection vaccine doses in a population that was first exposed to infection during a relatively short and well-defined time window. We further investigated whether the number of pre-infection vaccination doses modified the protective effect of the post-infection dose against Omicron BA.5 sublineage. To avoid ambiguity in the text, in the following sections, we often refer to vaccine doses given after the initial infection as “post-infection vaccination” or “post-infection vaccine doses”.

<https://doi.org/10.7554/eLife.94990.2.sa0>